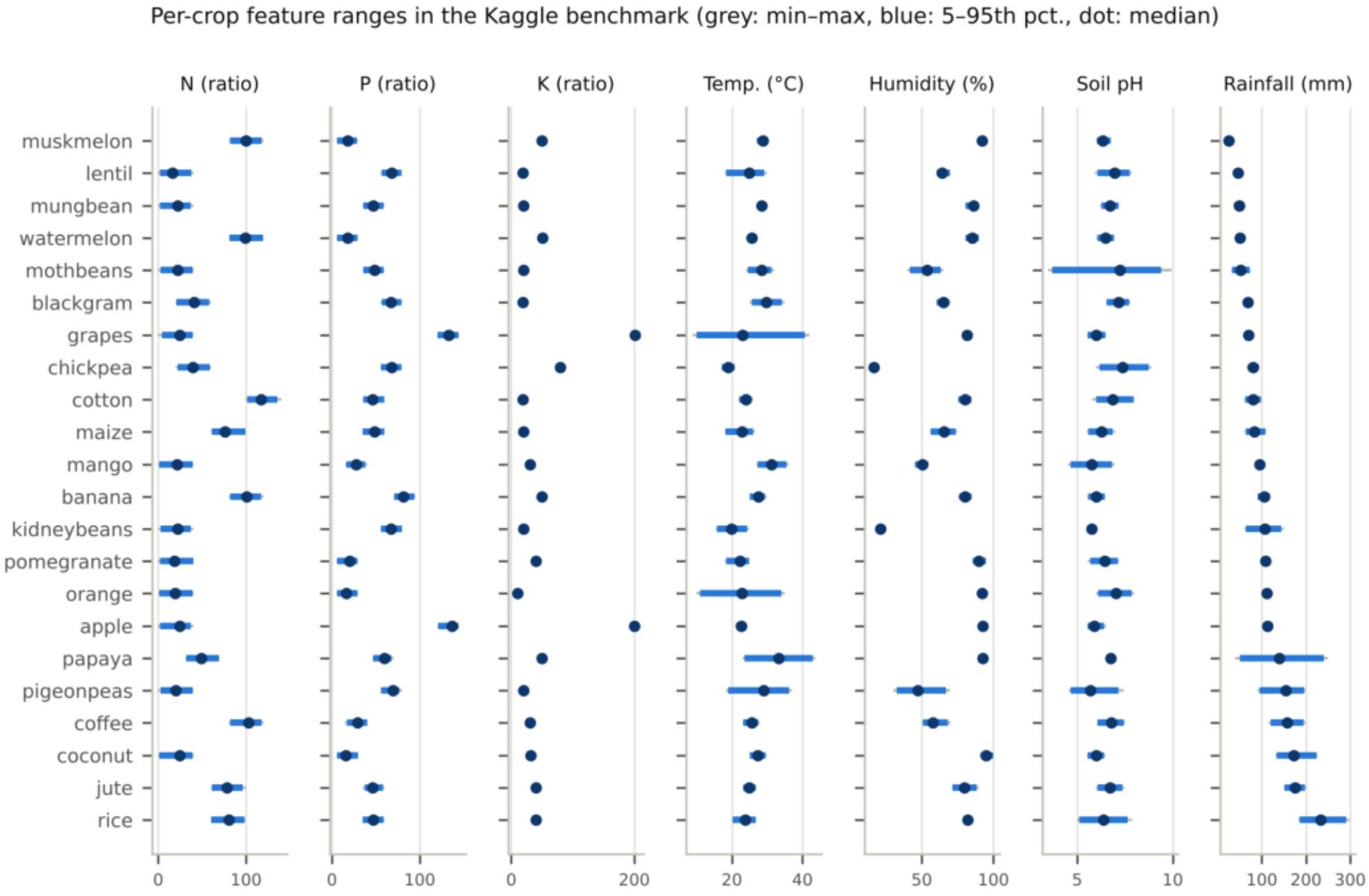


Fig 1. Per-crop feature ranges (Kaggle). Non-overlapping ranges make most crops separable by one or two features.

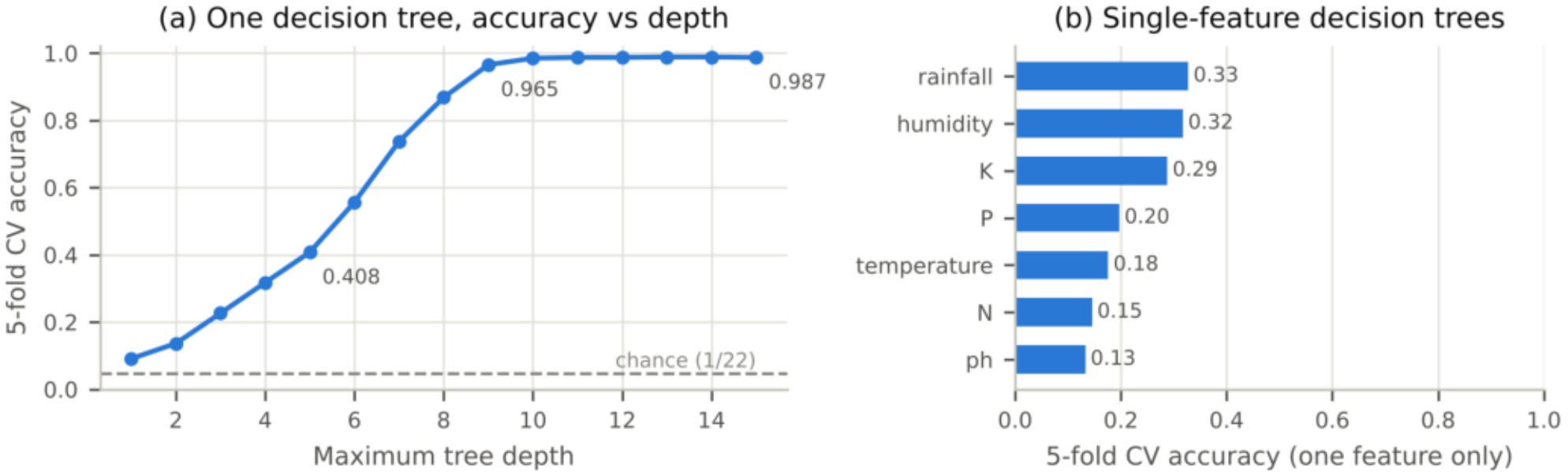


Fig 2. (a) One decision tree vs depth; (b) single-feature trees. Humidity+rainfall alone: 76.9% CV accuracy.

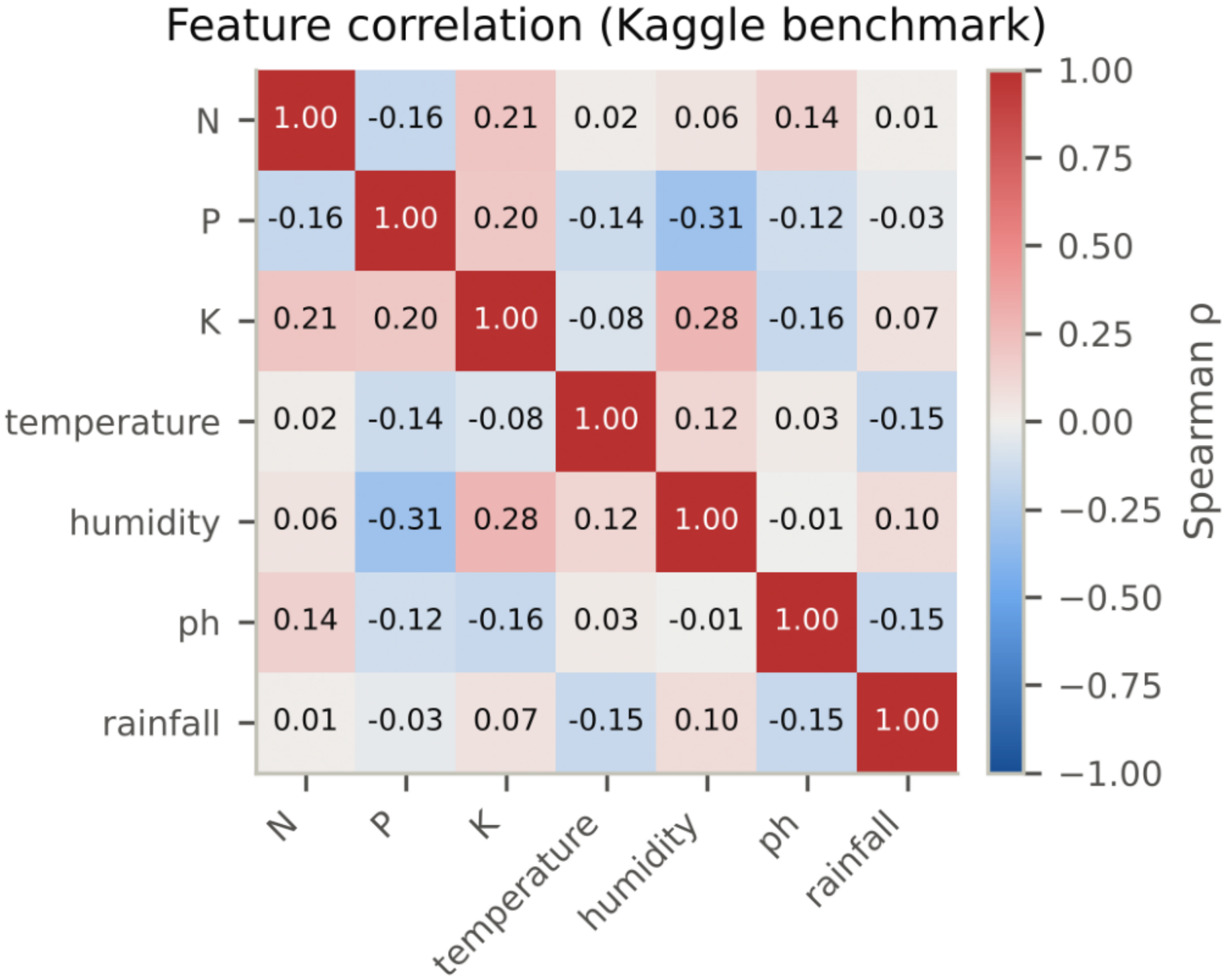
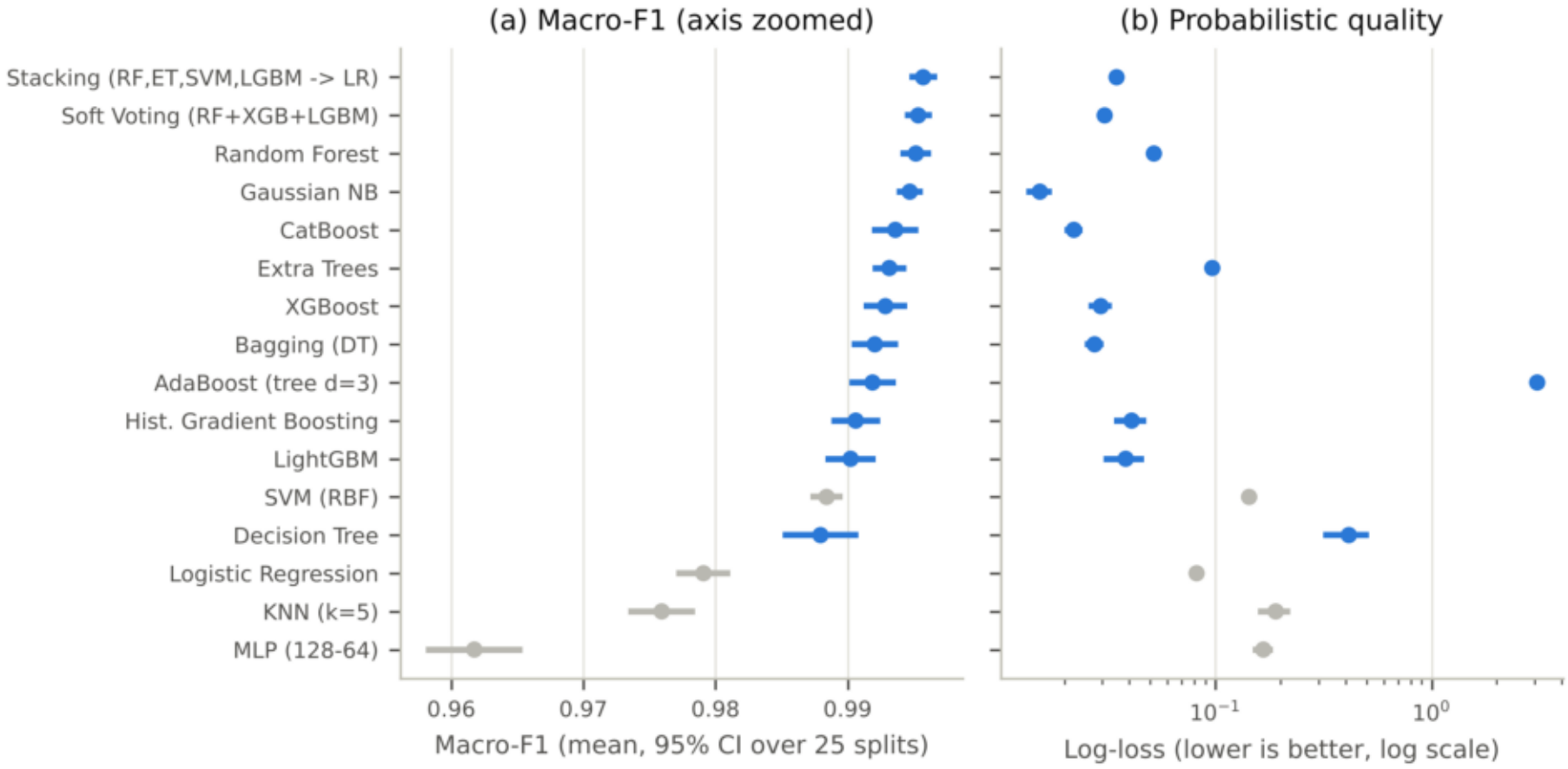


Fig 3. Spearman correlation between features.



Blue: best model (Stacking (RF,ET,SVM,LGBM -> LR)) and models not significantly worse (corrected resampled t-test, Holm-adjusted p ≥ 0.05). Grey: significantly worse.

Fig 4. 16 models, 5×5 repeated stratified CV. 11/16 not significantly worse than the best (Holm-corrected).

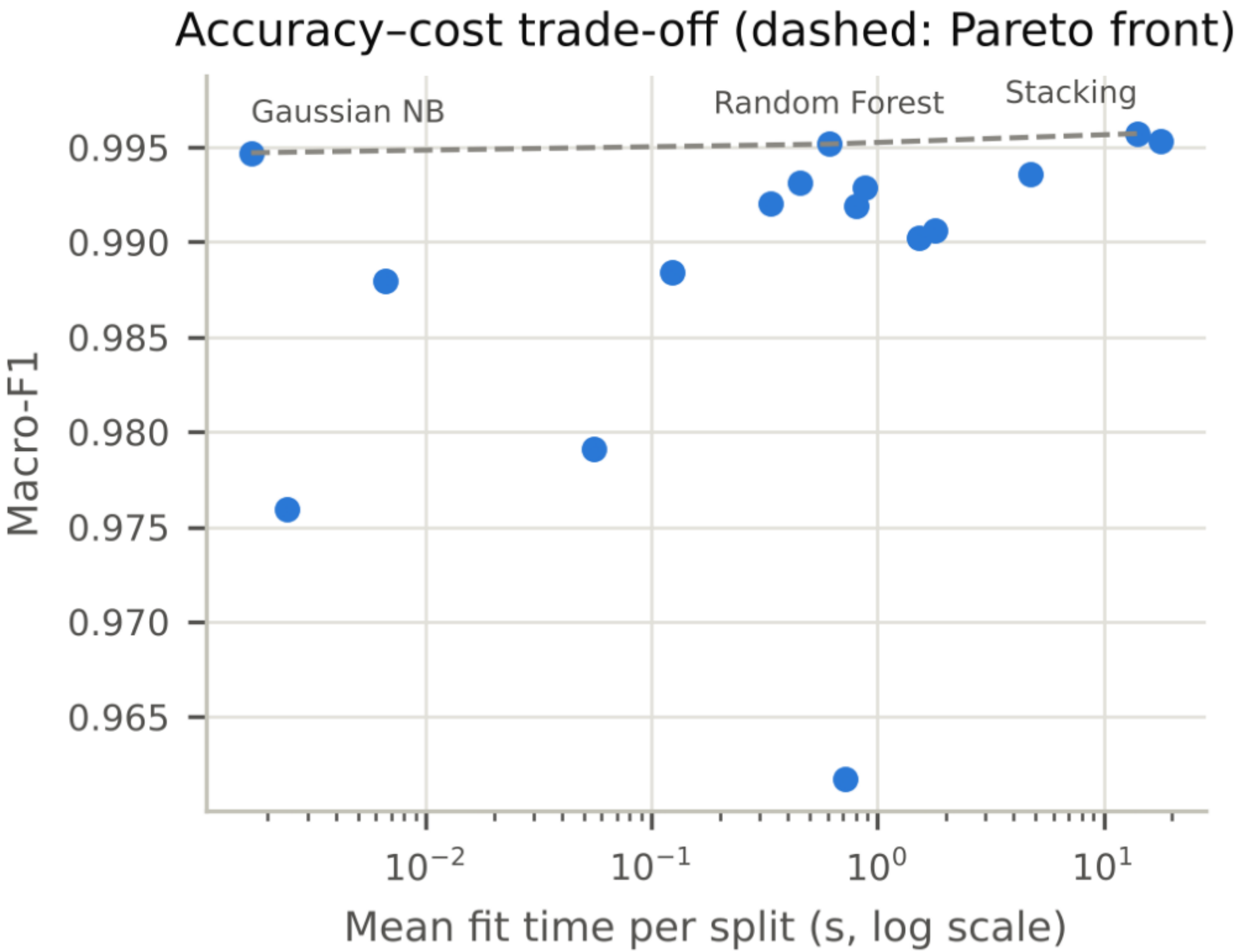


Fig 5. Fit time vs macro-F1; Gaussian NB is on the Pareto front at ~0 s.

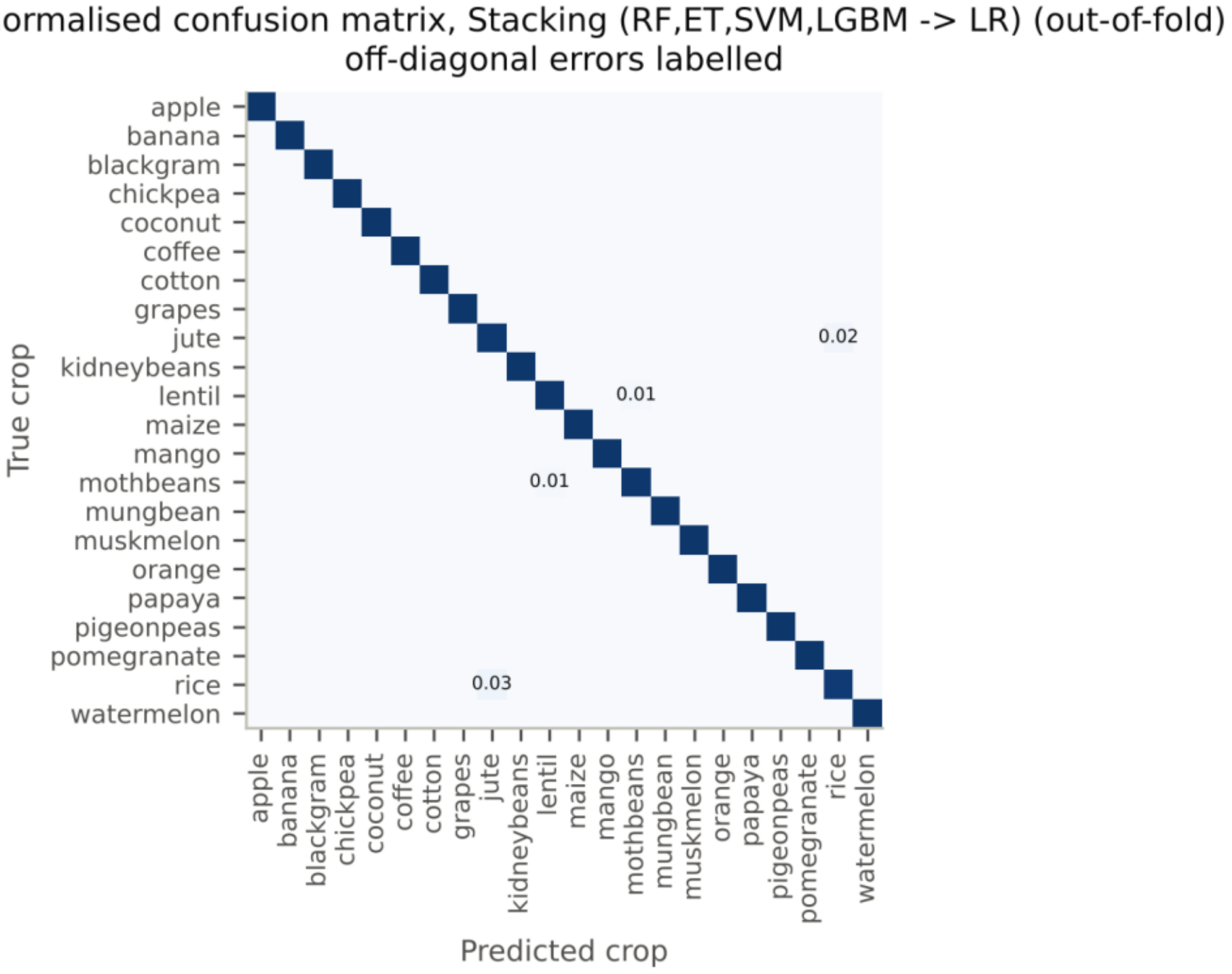


Fig 6. Confusion matrix of the best model; residual errors are rice/jute and lentil/mothbeans.

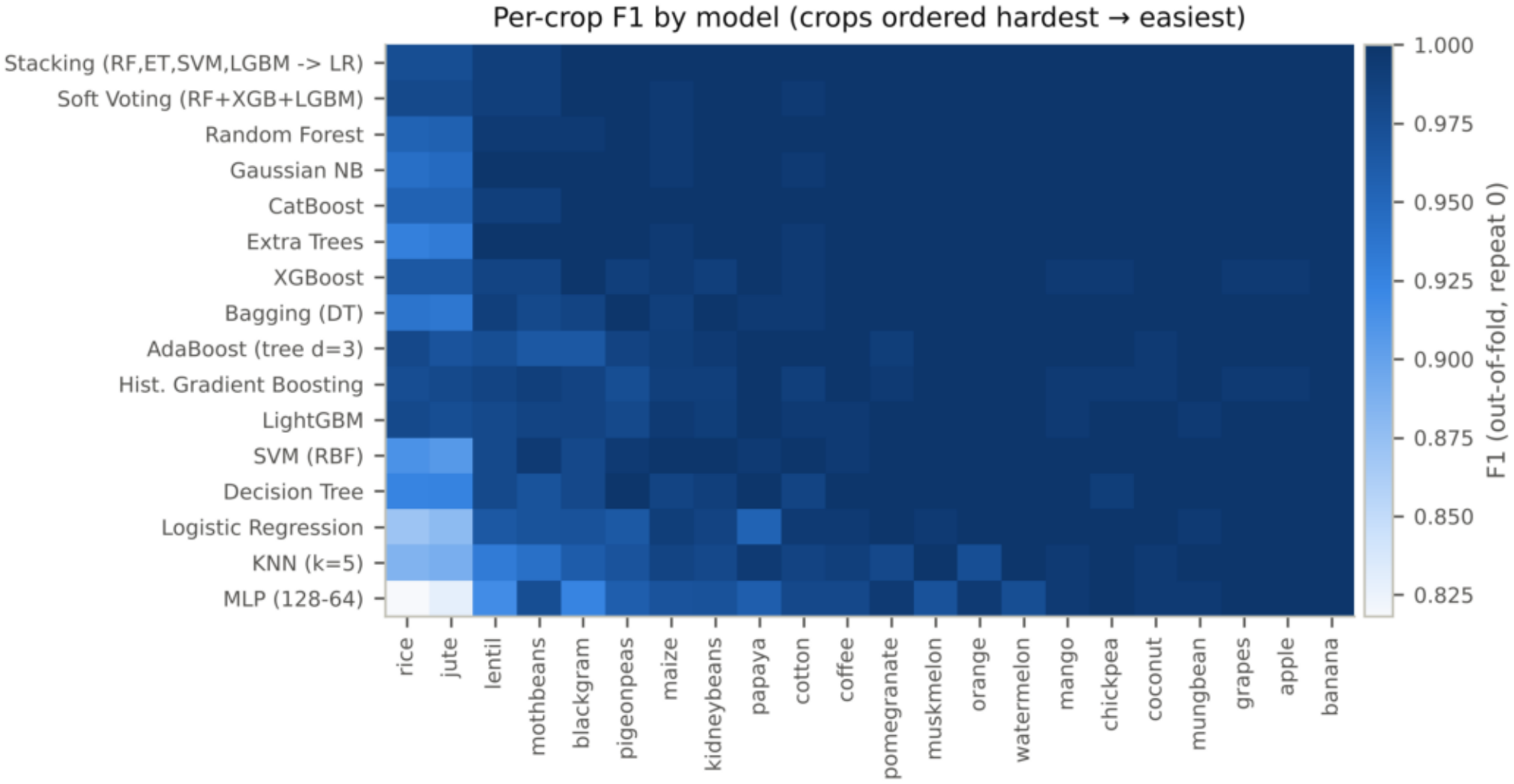


Fig 7. Per-crop F1 for every model.

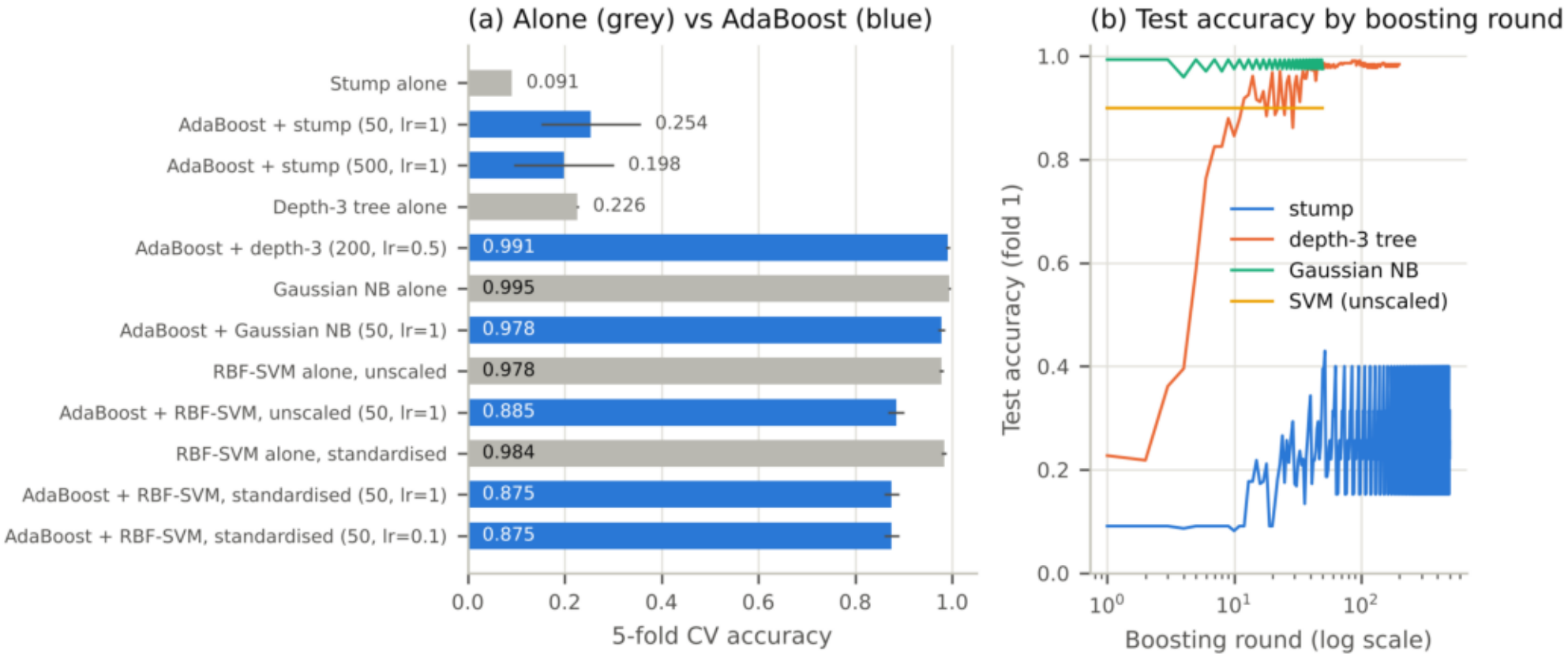


Fig 8. AdaBoost with default stumps collapses to 20-25%; boosting SVM/NB lowers their accuracy.

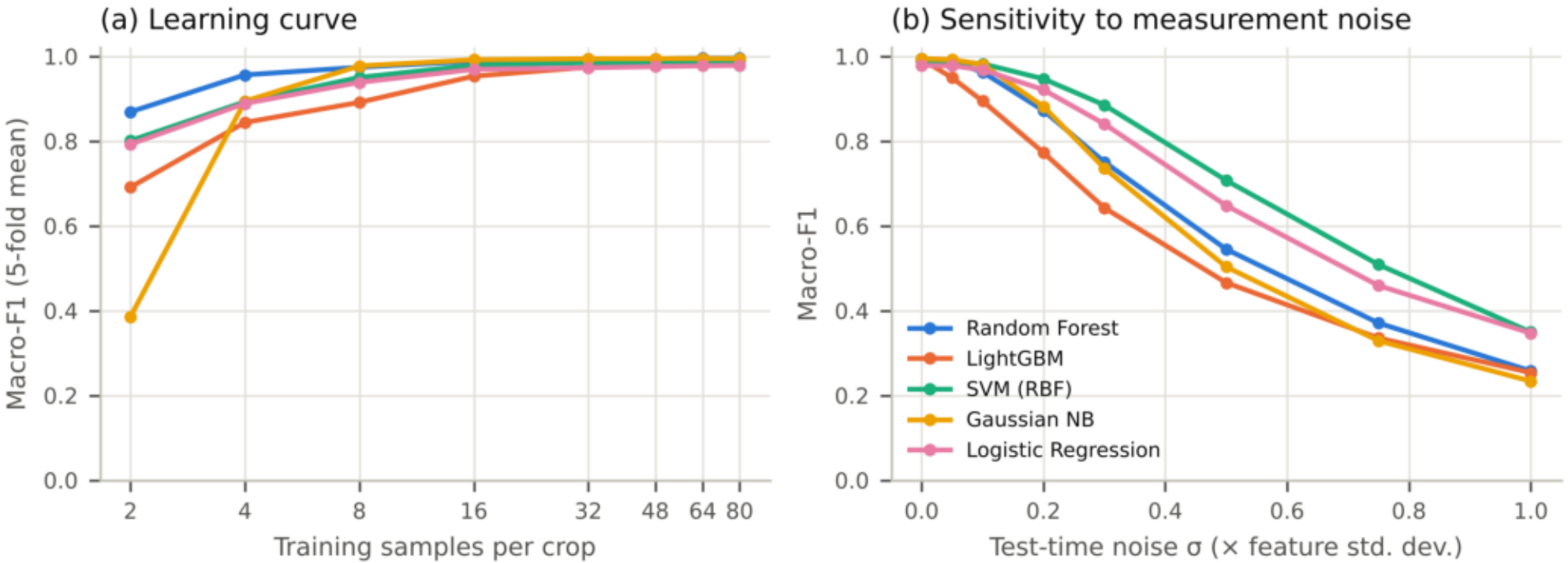


Fig 9. (a) RF reaches 0.97 macro-F1 with 8 samples per crop; (b) ranking changes under measurement noise.

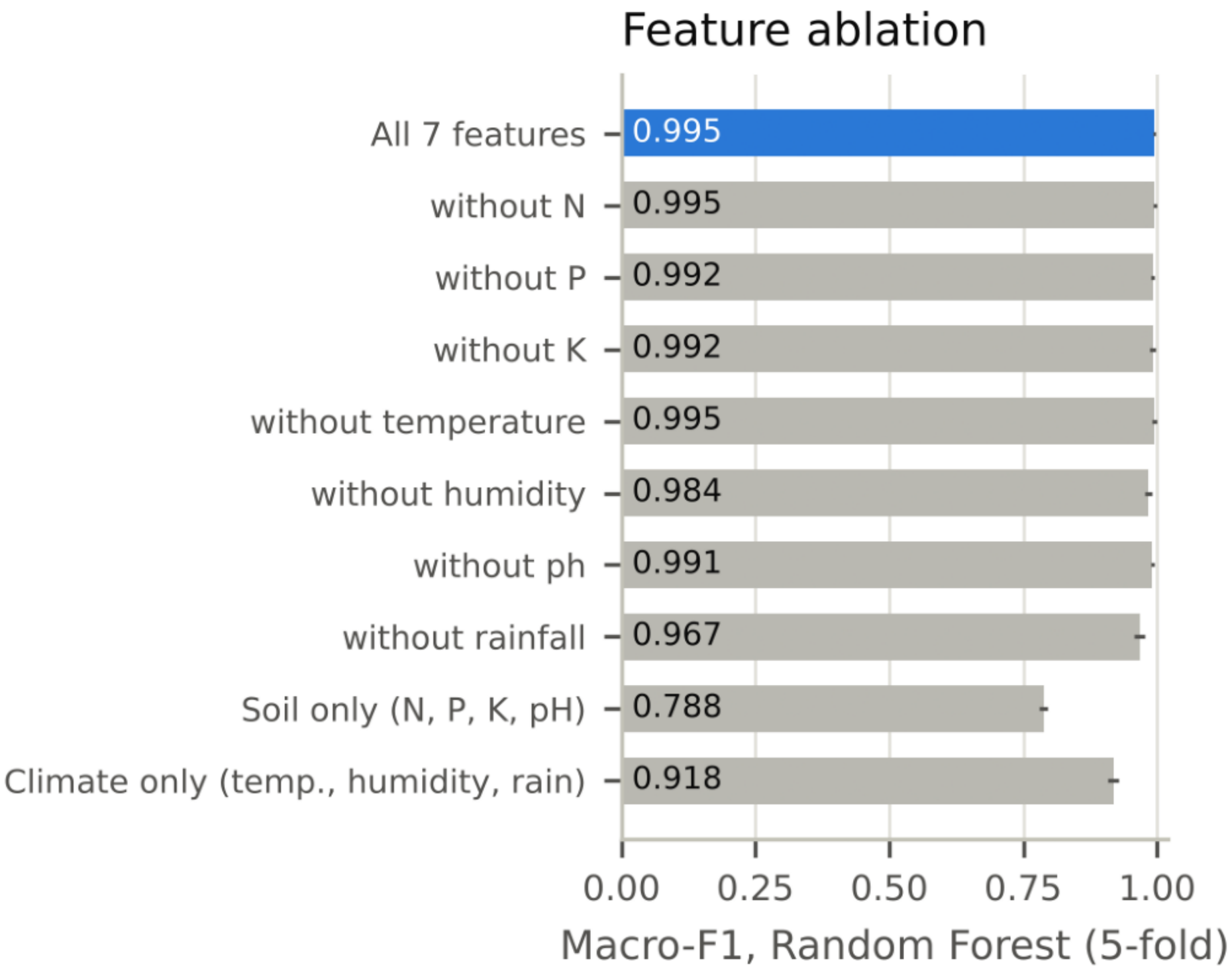


Fig 10. Climate-only features beat soil-only (0.918 vs 0.788 macro-F1).

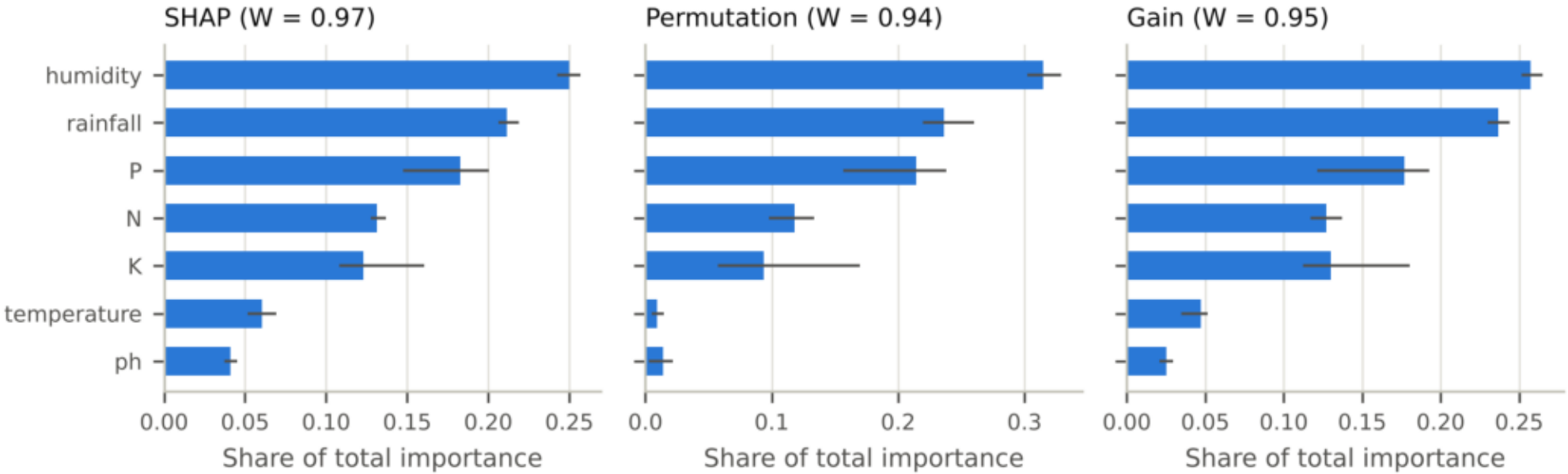


Fig 11. SHAP, permutation and gain importance with split ranges and Kendall's W.

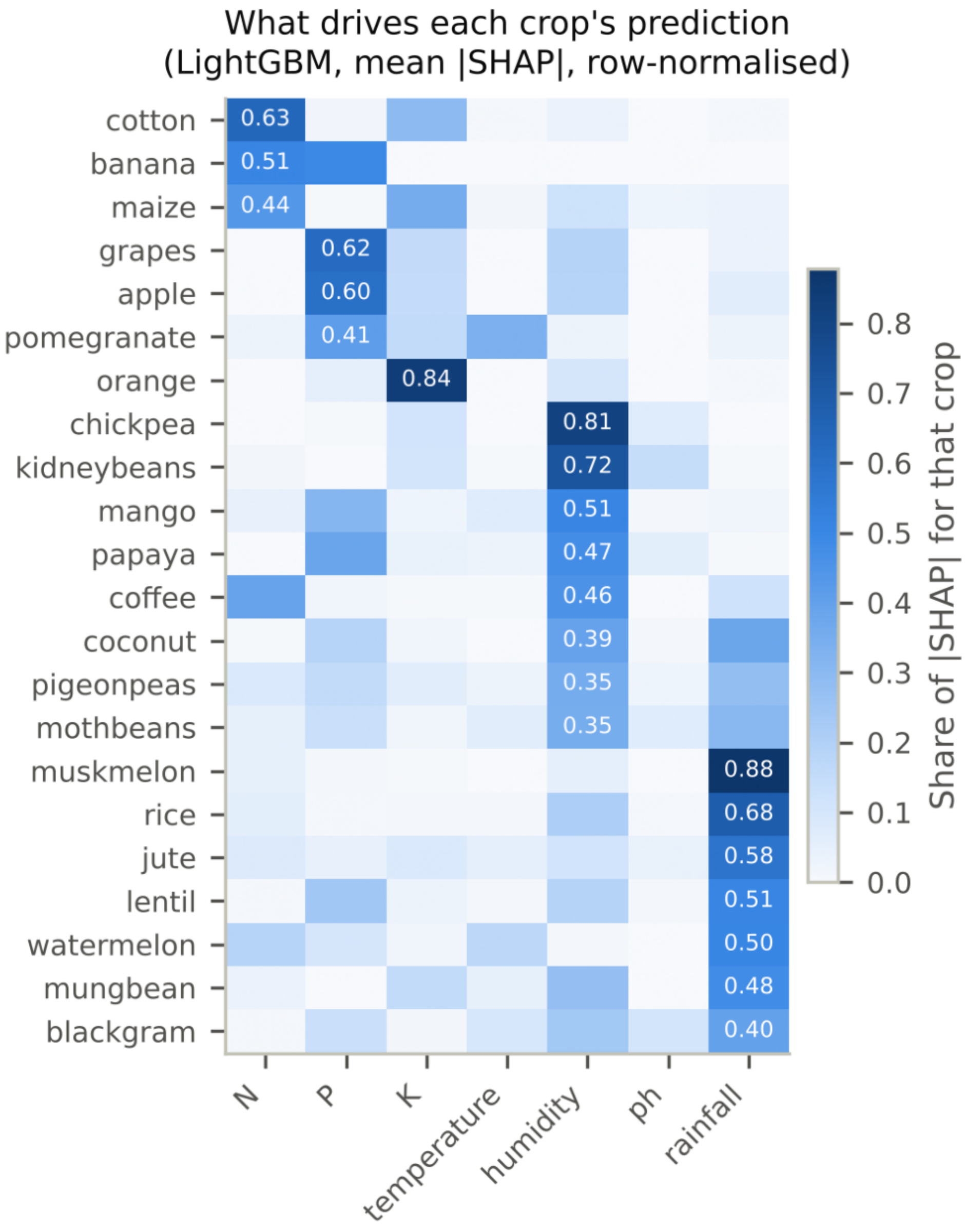


Fig 12. Feature driving each crop's prediction (row-normalised mean |SHAP|).

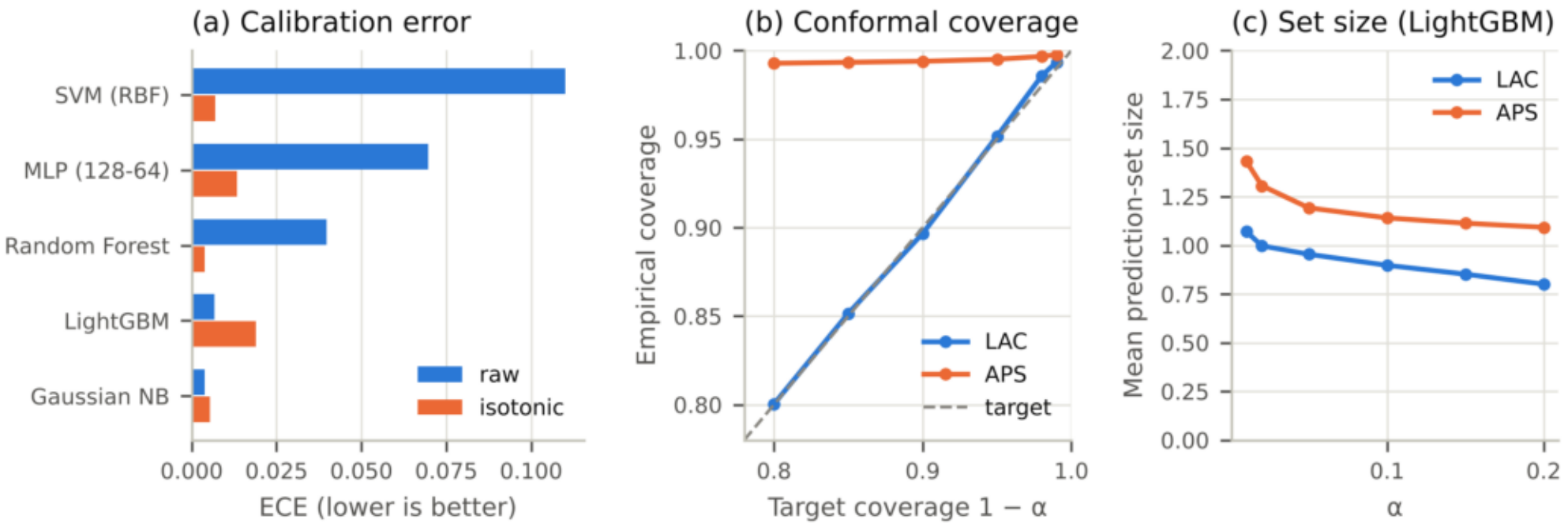


Fig 13. (a) ECE before/after isotonic; (b,c) conformal coverage and set size.

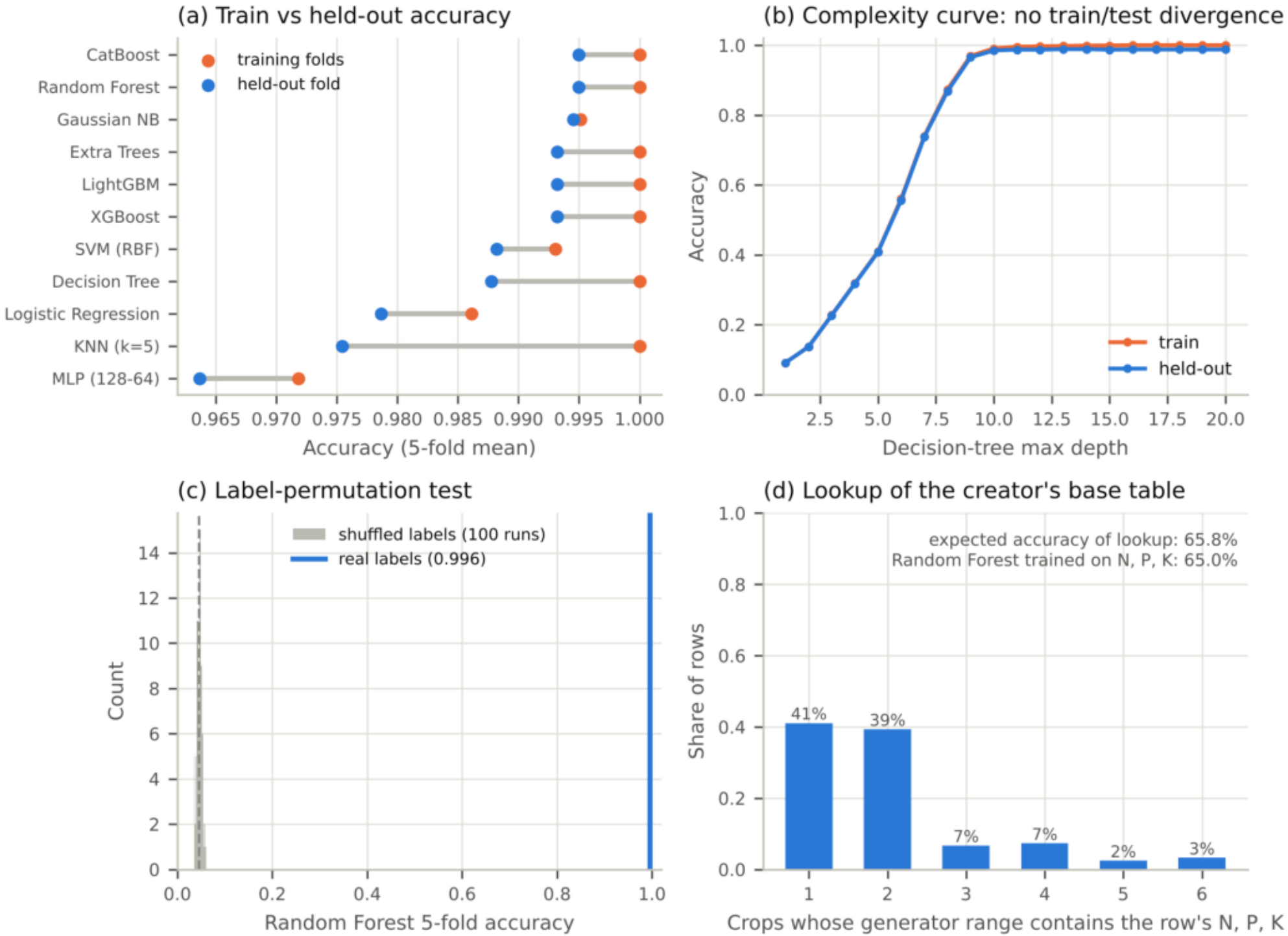


Fig 14. Not overfitting: small train/test gaps, no divergence, chance-level accuracy with shuffled labels. N,P,K are generated from the label: lookup of the creator's table 65.8%, RF on N,P,K 65.0%.

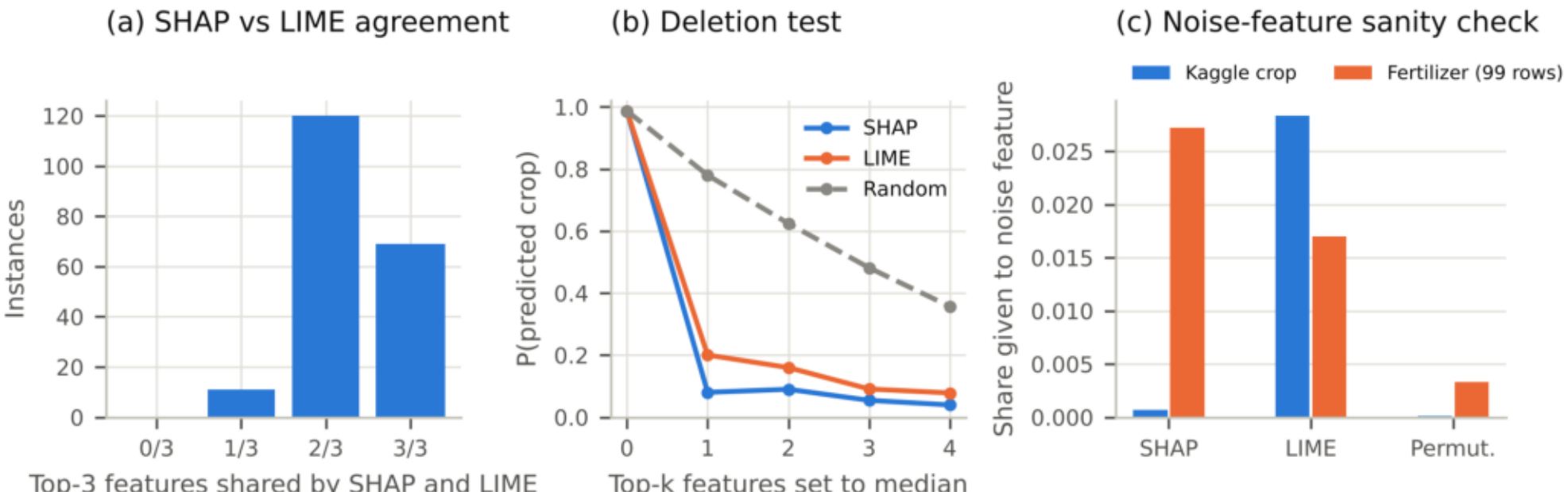


Fig 15. SHAP vs LIME agreement, deletion faithfulness, and importance given to a pure-noise feature.

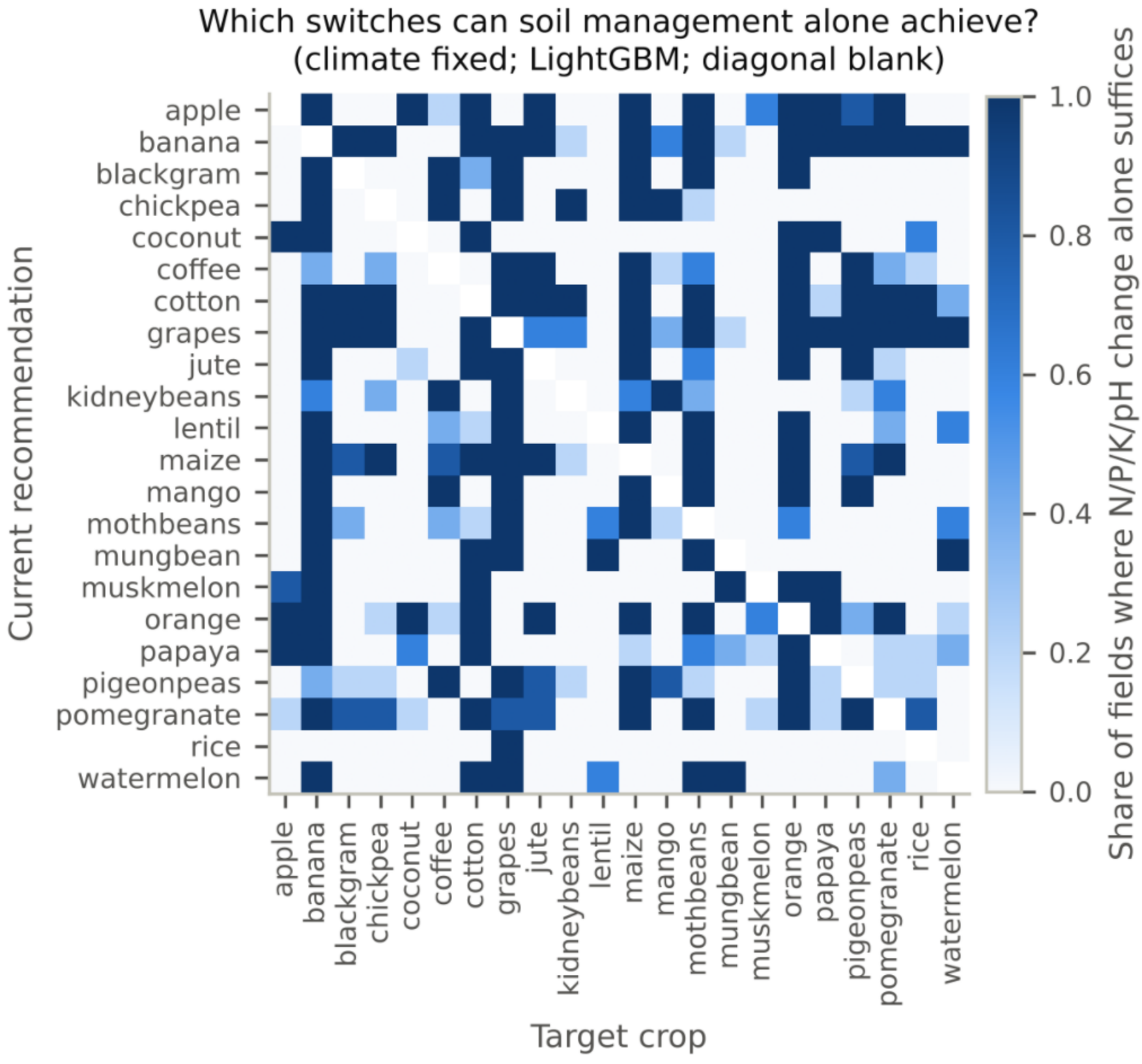
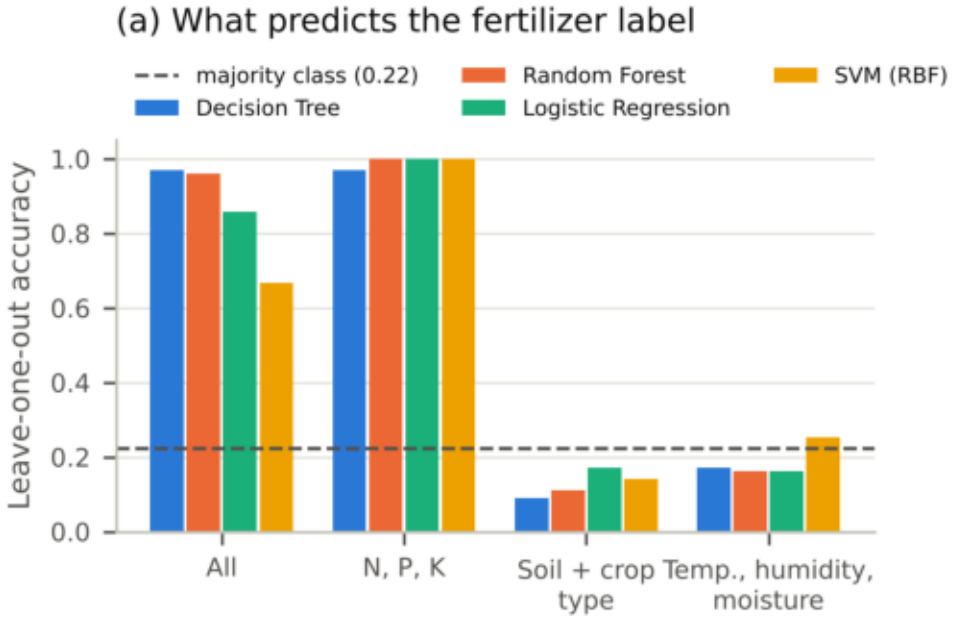


Fig 16. Crop switches reachable by changing N, P, K, pH only (climate fixed): 38% of pairs.



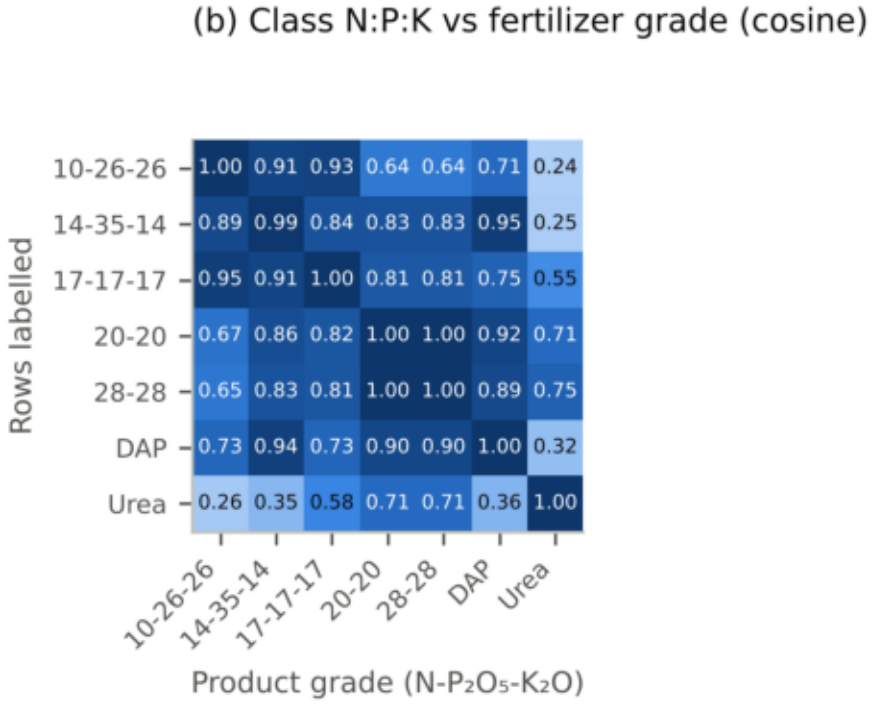


Fig 17. Fertilizer benchmark: N,P,K alone give 97-100% LOO accuracy; environment, soil and crop type are at or below the majority class; class N:P:K mirrors the fertilizer grade.